Practice workbook · Factor theorem: show a factor or find a constant

Substituting a single value into a polynomial to show that a given linear expression is a factor, to find one unknown constant, or to produce a relation between constants.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-08

1. Question 3 · part a · 2 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part a only.

3 Given that $(x - 4)$ is a factor of $px^3 - 31x^2 + 25x + 12$ where p is a constant,

(a) show that $p = 6$

(2)

(b) Solve the equation $6x^3 - 31x^2 + 25x + 12 = 0$

Show clear algebraic working.

(4)

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1. Question 3 · part a · 2 marks · Working and answer

Working and answer

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2. Question 9 · part b.i · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

9

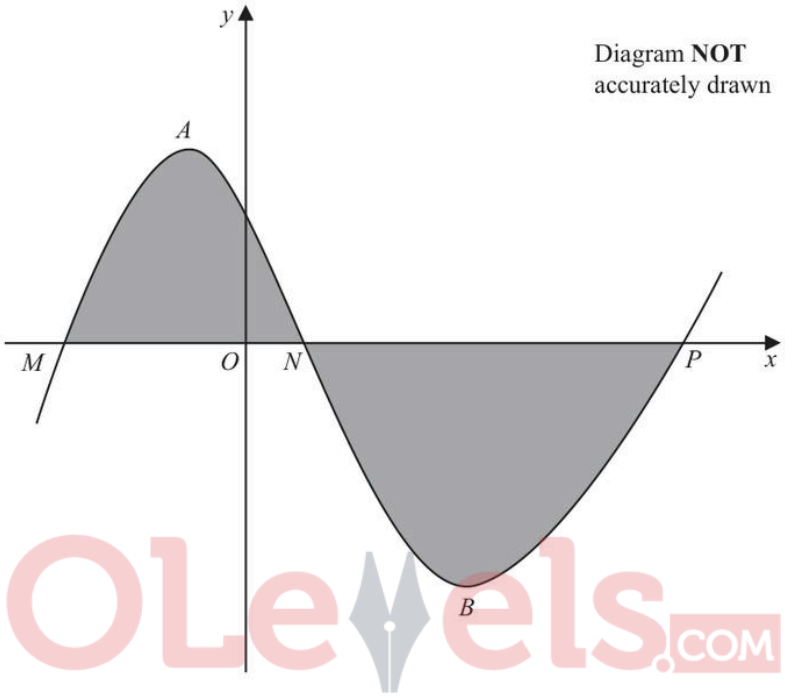


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^3 + ax^2 + bx + c$$

The curve C has a maximum at the point A with coordinates $\left(-\frac{1}{3}, \frac{100}{27}\right)$ and a minimum at the point B with coordinates $(2, -9)$

Given that a, b and c are integers

(a) show that $a = -5, b = -4$ and $c = 3$

(5)

(b) (i) Show that $(x+1)$ is a factor of $f(x)$

(1)

(ii) Hence, or otherwise, use algebra to factorise $f(x)$ completely.

(3)

The curve C crosses the x-axis at the points M, N and P
The finite regions shown shaded in Figure 4 are bounded by the curve C and parts of the x-axis from M to N and from N to P

(c) Use algebraic integration to determine the total area of the shaded regions. Give your answer as an exact fraction.

(4)

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P 7 2 8 6 5 A 0 2 8 3 6

2. Question 9 · part b.i · 1 mark · Working and answer

Working and answer

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3. Question 1 · part a · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

1 $f(x) = (x - 3)[x^2 + (p - 2)x + q]$

Given that $f(0) = -12$

(a) find the value of q.

(2)

(b) Find the range of values of p for which the cubic equation $f(x) = 0$ has only one real root.

(5)

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(Total for Question 1 is 7 marks)

3. Question 1 · part a · 2 marks · Working and answer

Working and answer

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4. Question 2 · part b · 1 mark

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

2 $f(x) = x^3 + px + q$ where p and q are constants.

The remainder when $f(x)$ is divided by $(x - 1)$ is -12

The remainder when $f(x)$ is divided by $(x - 4)$ is 30

(a) Find the value of p and the value of q.

(6)

Using your values of p and q

(b) show that $f(3) = 0$

(1)

(c) Express $f(x)$ as a product of linear factors.

(3)

(d) Hence solve the equation $f(x) = 0$

(1)

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4. Question 2 · part b · 1 mark · Working and answer

Working and answer

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5. Question 10 · part a · 1 mark

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part a only.

10 $f(x) = 32x^3 - 33x + 1$

(a) Show that $f(1) = 0$

(1)

(b) Hence using an algebraic method solve $f(x) = 0$

(4)

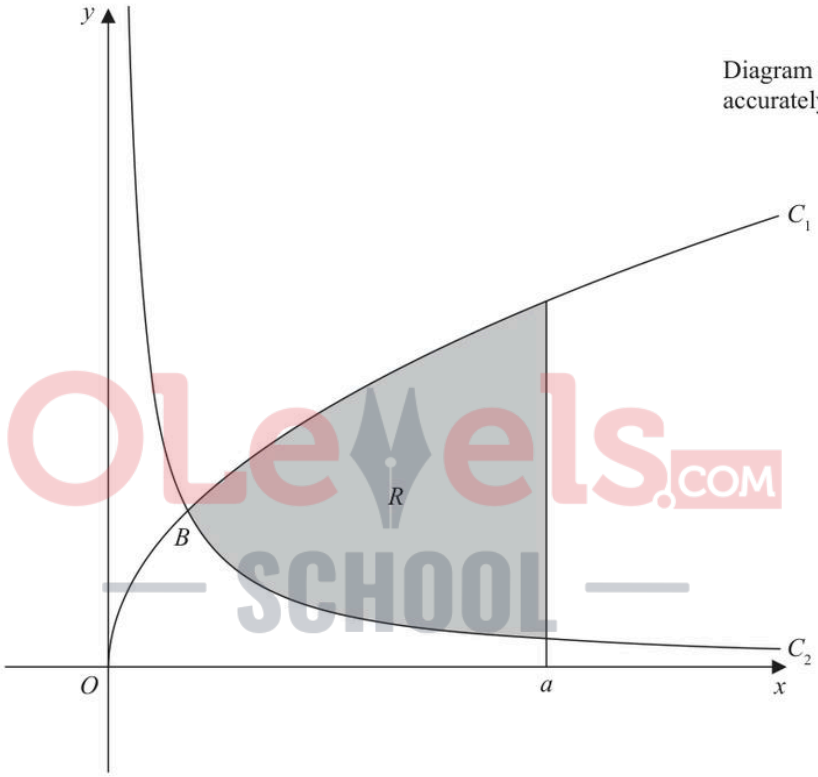


Figure 4

The region R, shown shaded in Figure 4, is bounded by the curve C_1 with equation $y = \sqrt{x}$, by the curve C_2 with equation $y = \frac{1}{8x}$ and by the line with equation $x = a$

The curves C_1 and C_2 intersect at the point B, with x coordinate p, where $p < a$

(c) Find the value of p.

(2)

The region R is rotated through 360° about the x-axis to generate a solid with volume $\frac{27\pi}{64}$

(d) Use algebraic integration to find the value of a.

(7)

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5. Question 10 · part a · 1 mark · Working and answer

Working and answer

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6. Question 6 · part a · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

6 $f(x) = x^3 + (p + 1)x^2 - 10x + q$

where p and q are integers.

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that $9p + q + 6 = 0$

(3)

Given that $(x + p)$, where $p > 0$, is also a factor of $f(x)$

(b) show that $p^2 + 10p + q = 0$

(3)

(c) Hence find the value of p and the value of q.

(5)

(d) Using your values of p and q, factorise $f(x)$ completely.

(2)

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6. Question 6 · part a · 3 marks · Working and answer

Working and answer

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7. Question 6 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

6 $f(x) = x^3 + (p + 1)x^2 - 10x + q$

where p and q are integers.

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that $9p + q + 6 = 0$

(3)

Given that $(x + p)$, where $p > 0$, is also a factor of $f(x)$

(b) show that $p^2 + 10p + q = 0$

(3)

(c) Hence find the value of p and the value of q.

(5)

(d) Using your values of p and q, factorise $f(x)$ completely.

(2)

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7. Question 6 · part b · 3 marks · Working and answer

Working and answer

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8. Question 1 · part a · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

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1 $f(x) = 6x^3 - 13x^2 + ax - 10$ where a is a constant

Given that $(3x - 2)$ is a factor of $f(x)$

(a) show that $a = 21$

(2)

(b) Hence show algebraically that the curve $y = f(x)$ has only one intersection with the x-axis.

(4)

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(Total for Question 1 is 6 marks)

8. Question 1 · part a · 2 marks · Working and answer

Working and answer

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9. Question 7 · part a · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part a only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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9. Question 7 · part a · 2 marks · Working and answer

Working and answer

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10. Question 10 · part a · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

10 (a) Use the factor theorem to show that $(4x - 3)$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Using formulae given on page 2, show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

(c) Show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

becomes the equation

$$16x^3 + 11x - 15 = 0$$

by using the substitution $x = \cos\theta$

(4)

(d) Hence show that any solution of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

is given by $\cos\theta = \frac{3}{4}$

(4)

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10. Question 10 · part a · 2 marks · Working and answer

Working and answer

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